

1a) If the user is allowed to put any system call (as in, outside the syscall table) the kernel will add it to the sys call table which allows an attacker to jump to any memory address, essentially gaining control/access to the computer.

1b1) Yes, because the program was running in user mode when the interrupt arrived.

1b2) No, because syscalls are handled in kernel mode, so we would already be in kernel mode: the TSS doesn't care.

1b3) No, because interprocessor interrupts always run in kernel mode.

1c) The system will halt with a kernel panic, resulting either in a bootloop or a waiting state where the system cannot do anything.

1d) The flag exists so that the kernel's scheduler can either be swapped out for another task that is more efficient; in summary any task that needs the CPU more can grab the CPU.